Sisir Kausthubh

Robotics & Edge-AI Engineer

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SUMMARY

I make robot foundation models fast and frugal enough for real robots without losing accuracy, and prove it closed loop. My path runs from perception (computer vision) through decisions (VLMs, agents) to action (robot policies), aimed at frontier robotics: reliable robot brains running on the robot. Every result reports speed, accuracy and efficiency together, against a published baseline.

PROJECTS: ROBOTICS

SmolVLA fast path on LIBERO | independent research Oct 2026 - Present

- **7.7x faster per decision, no retraining:** 101 ms to 13 ms median, 120 ms to 16 ms p99; **9.3x less energy** (10.5 J to 1.1 J) on an RTX 5090 laptop GPU.
- **Task success held:** 83.00% vs 83.75% over the same 400 closed-loop episodes (paired McNemar test, no significant change). Trade-off reported: 71 to 89 s cold startup.
- How: whole-model torch.compile, 1 denoising step instead of 10, batched camera encoding, unused camera slot removed. Tested TensorRT, INT8 and FP8; rejected FP8 vision after success fell to 80%.

Warehouse robot navigation with deep RL BTech

- DDPG continuous-control policy for reaching goals while avoiding obstacles in a simulated Gazebo warehouse; plus TD3 (CNN features) and Q-learning navigation experiments.

Smart home and security system | Raspberry Pi + Arduino BTech

- Live Pi-camera stream (640x480, 24 fps), PIR motion sensor that snaps and emails intruder photos, and a Bluetooth serial door-control prototype on Arduino.

OPEN SOURCE & TALKS

- **Hugging Face LeRobot:** reviewed and benchmarked PR #4634 (SmolVLA KV cache + compile): measured 2.2x end to end vs 4.2x claimed. SmolVLA camera-batching change in progress.
- **Speaker:** lightning talk on the SmolVLA results, NVIDIA Jetson AI Lab community meeting, 10 Nov 2026.

PROJECTS: COMPUTER VISION

- **visual-product-tagging:** fine-tuned a 4B vision-language model in 48 minutes on one GPU; 85.3% on ImplicitAVE (GPT-4V: 86.8%) at INR 0.67 per 1,000 products.
- **document-extraction-pipeline:** receipts and GST invoices to checked spreadsheet rows; 84.0 field F1 on CORD (Donut: 83.3 on the same scorer), with business-rule checks and human review.
- **Traffic-sign recognition (GTSRB, 43 classes):** CNN with augmentation compared against a fully connected network, with per-class evaluation.
- **Real-time video style transfer:** feed-forward residual network with VGG16 perceptual loss; foreground and background of a video stylized separately.

PROJECTS: AI AGENTS & GENAI | each runs on one laptop GPU

- **catalog-automation-agent** (LangGraph, MCP server, n8n): auto-tagged 79.2% of 1,610 products at 94.6% accuracy; crash test lost 0 items with 0 duplicates; 45 products/s.
- **docs-rag-assistant:** page-cited RAG with a local 4-bit Gemma 4 26B; 61.8% on FinanceBench (paper's GPT-4-Turbo RAG: 19%).
- **LLM fine-tuning:** Llama 3.2 3B with QLoRA (Unsloth, TRL), trained on assistant replies only, exported to GGUF for local use.
- **AI short-video generator** (Next.js, Remotion): topic to script, generated images, narration and word-timed captions in one workflow.

PROJECTS: LANGUAGE & SPEECH AI | BTech research

- **Tamil-to-Telugu speech-to-speech translation:** Whisper speech recognition (best of 4 models compared on Common Voice 11), IndicTrans2 translation, VITS/MMS speech synthesis.
- **Tamil multimodal hate-speech detection** (final-year research): IndicBERT text and ViT/ResNet video features. **Malayalam headline generation** with a custom encoder-decoder Transformer; Indic document OCR tooling.

PLANNED PROJECTS | designed, starting Nov 2026

GR00T N1.7 fast-path audit (TensorRT/FP8, closed loop, RTX and Jetson) | SmolVLA over ROS 2 | fault-tolerant agent and robot swarms in Isaac Lab | distilling large robot models for Jetson.

SKILLS

Inference & ML: Python, PyTorch, torch.compile, TensorRT, INT8/FP8 quantization, vLLM, QLoRA/LoRA fine-tuning, Hugging Face (LeRobot, Transformers, TRL)
Robotics & sim: VLAs (SmolVLA), LIBERO, MuJoCo, Gazebo, deep RL (DDPG, TD3), Raspberry Pi, Arduino | learning ROS 2, Isaac Lab, Jetson
Vision, agents & GenAI: vision-language models, CNNs, LangGraph, MCP, RAG, Whisper, paired-statistics evaluation | **Tools:** Linux, Git, uv, Next.js

EDUCATION

BTech, Computer Science and Engineering (AI) | Amrita Vishwa Vidyapeetham Sep 2021 - Jun 2025